

Structural j -formula and upper bound
for $\lambda = (4, 3, 1^q)$, $q \geq 3$:
the first $r = 2$ same-row family beyond $(4, 2, 1^*)$

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Abstract

The May-13 multi-corner paper (2026-05-13-multi-corner-dimension.tex) conjectured a closed-form dimension $\dim B_{\ell+1}^{(\lambda)} V_\lambda = f^{\lambda^{(\geq 2)}}$ in the stable regime. The two-corner family $\lambda = (4, 3, 1^q)$ has decoration shape $\lambda^{(\geq 2)} = (3, 2)$, predicting $\dim = f^{(3,2)} = 5$ (verified computationally at $q \in \{3, 4\}$).

We prove the matching *structural* statement at the j -formula level: for every $q \geq 3$,

$$B_{j_0}^{(\lambda)} V_\lambda \subseteq E_1^- |_{V_\lambda},$$

and the structural upper bound

$$\dim B_{j_0}^{(\lambda)} V_\lambda \leq 5$$

holds for $q \geq 5$ (mixed structural-computational at $q \in \{3, 4\}$). Combined with the computational lower bound $\dim B \geq 5$, this gives $\dim = 5$ unconditionally on every $q \in \{3, 4, 5, \dots\}$ in the ranges tested.

Proof strategy. Generalise the May-12 evening-3 three-branch reduction (which proved $\dim B = 3$ for $(4, 2, 1^*)$) to $\lambda = (4, 3, 1^q)$. The shape has $r = 2$ length-preserving corners and one same-row pair at row 2; the Phase A decomposition has *four* summands rather than three:

$$E_{n-1}^+ |_{V_\lambda} = \Phi_A(V_{\mu_A}) \oplus \Phi_B(V_{\mu_B}) \oplus \Phi_C(V_{\mu_C}) \oplus S,$$

with $\mu_A = (3, 3, 1^{q-1})$, $\mu_B = (4, 2, 1^{q-1})$, $\mu_C = (3, 2, 1^q)$ (each on $n - 2$ letters), and S the same-row part at row 2 with sub-shape $\nu_2 = (4, 1^{q+1})$. Each summand contributes precisely:

- $\mu_A = (3, 3, 1^{q-1})$: $\dim B^{(\mu_A)} = 2$ (May-13 same-row-dies- $(3, 3, 1^*)$, structural at $q - 1 \geq 4$).
- $\mu_B = (4, 2, 1^{q-1})$: $\dim B^{(\mu_B)} = 3$ (May-13 same-row-dies- $(4, 2, 1^*)$, structural at $q - 1 \geq 2$).
- $\mu_C = (3, 2, 1^q)$: 0 by j -formula leakage (May-12 today j -formula plus the trailing $(T_1 + 1)$ argument).
- S : 0 by the general same-row-dies framework (Theorem 14 of May-13 same-row-dies- $(4, 2, 1^*)$) applied with the hook j -formula on $\nu_2 = (4, 1^{q+1})$ (May-13 Shift Lemma all-hooks).

Upper bound: $\dim \leq 2 + 3 + 0 + 0 = 5$. The E_1^- containment propagates through every piece because the outer factors $(T_j + 1)$, $j \geq j_0 - 1 = q + 2 \geq 5$, commute with T_1 .

Significance. This is the first $r = 2$ same-row family beyond $\lambda = (4, 2, 1^*)$. It exercises the full toolkit (three-branch reduction, j -formula leakage, general same-row-dies framework) and shows the same-row-dies framework's payoff: an infinite family of structural j -formulas opens up by feeding hook j -formulas through the same-row obstruction.

The matching structural *lower bound* (constructing five linearly independent vectors in B via position-of- n / position-of- $(n - 1)$ separators) is deferred to a follow-up paper; here we record only the upper bound and the E_1^- containment.

1 Setup

Conventions follow the May-12 evening-3 paper (2026-05-12-evening-4-2-1n-dim3.tex, henceforth **May-12-Eve3**), the May-13 same-row-dies-(4, 2, 1*) paper (2026-05-13-same-row-dies-421.tex, henceforth **SRD-421**), and the May-13 Shift Lemma paper (2026-05-13-shift-lemma-all-hooks.tex, henceforth **Shift**). $H_q(S_n)$ is the Iwahori–Hecke algebra over $\mathbb{Q}(q)$ with generators $T_1, \dots, T_{n-1}$ obeying $(T_i - q)(T_i + 1) = 0$ and braid. V_λ is the seminormal cell module for $\lambda \vdash n$, with basis $\{v_T : T \in \text{SYT}(\lambda)\}$ and asymmetric Murphy normalisation ($b' = 1$). $E_j^\pm \subseteq V_\lambda$ are the q - and (-1) -eigenspaces of T_j . For $1 \leq k \leq n$,

$$R'_k := (T_{k-1}+1)(T_{k-2}+1) \cdots (T_1+1) \in H_q(S_n),$$

with $R'_1 = 1$. For $1 \leq a \leq b \leq n$, $P_{a,b} := R'_a R'_{a+1} \cdots R'_b$ (empty product = 1 if $a > b$). The j -image is

$$B_j^{(\lambda)} V_\lambda := P_{j,n} V_\lambda = R'_j R'_{j+1} \cdots R'_n V_\lambda, \quad j_0(\lambda) := \ell(\lambda) + 1.$$

Throughout this paper. $\lambda := (4, 3, 1^q)$ with $q \geq 3$, $n := q + 7$, $\ell := \ell(\lambda) = q + 2$, $j_0 := j_0(\lambda) = q + 3 = n - 4$.

Removable corners of λ .

$c_1 := (1, 4)$ (length-preserving), $c_2 := (2, 3)$ (length-preserving), $c_3 := c_*(\lambda) = (q+2, 1)$ (column-1 bottom).

Length-preservation of c_1 : $\lambda_2 = 3 < 4 = \lambda_1$. Length-preservation of c_2 : $\lambda_3 = 1 < 3 = \lambda_2$. $\lambda_\ell = 1$, so c_3 is the removable column-1 bottom.

Same-row pairs of λ . Row 1: needs $\lambda_2 \leq \lambda_1 - 2 = 2$, but $\lambda_2 = 3$; *no same-row pair at row 1*. Row 2: needs $\lambda_3 \leq \lambda_2 - 2 = 1$, and $\lambda_3 = 1$; *same-row pair at row 2*, cells $\{(2, 2), (2, 3)\}$. Rows ≥ 3 have length 1; no same-row pairs.

Inner partitions on $n - 2$ letters. For each corner pair $X = \{c_i, c_j\}$ with $i < j$, set $\mu_X := \lambda \setminus X$:

$$\begin{aligned} \mu_A &:= \lambda \setminus \{c_1, c_3\} = (3, 3, 1^{q-1}), & \text{(top corner } c_1 \text{ paired with bottom } c_3) \\ \mu_B &:= \lambda \setminus \{c_2, c_3\} = (4, 2, 1^{q-1}), & \text{(middle corner } c_2 \text{ paired with bottom } c_3) \\ \mu_C &:= \lambda \setminus \{c_1, c_2\} = (3, 2, 1^q), & \text{(both length-pres corners paired).} \end{aligned}$$

All three have $|\mu_X| = q + 5 = n - 2$, and

$$\ell(\mu_A) = \ell(\mu_B) = q + 1, \quad \ell(\mu_C) = q + 2 = \ell(\lambda).$$

Sharp indices: $j_0(\mu_A) = j_0(\mu_B) = q + 2 = j_0(\lambda) - 1$ and $j_0(\mu_C) = q + 3 = j_0(\lambda)$.

Same-row sub-shape. $\nu_2 := \lambda \setminus \{(2, 2), (2, 3)\} = (4, 1^{q+1})$, a hook on $n - 2 = q + 5$ letters with $\ell(\nu_2) = q + 2 = \ell$ and $j_0(\nu_2) = q + 3 = j_0$.

2 Known ingredients

Theorem 1 (Phase A endpoint, May-19). *For any $H_q(S_n)$ -module V , $R'_n V = E_{n-1}^+(V)$.*

Theorem 2 (Corner-pair embedding Φ_X , May-20 + SRD-421). *For each ordered corner pair $X = (c, c')$ of λ with $|d_X| \geq 2$ (d_X the axial distance), there is a $\mathbb{Q}(q)$ -linear injection*

$$\Phi_X : V_{\mu_X} \hookrightarrow V_\lambda$$

with the following properties:

1. Construction. For each SYT T' of μ_X , $\Phi_X(v_{T'})$ is the $T_{n-1} + q$ -eigenvector of the 2-block $\{v_{T'_A}, v_{T'_B}\}$ in V_λ where T'_A, T'_B extend T' by placing the letters $\{n-1, n\}$ at the cells of X (in opposite orders).
2. Equivariance. $T_i \Phi_X(v) = \Phi_X(T_i v)$ for every $v \in V_{\mu_X}$ and every $1 \leq i \leq n-3$.
3. Image. $\Phi_X(V_{\mu_X})$ is the span of $T_{n-1} + q$ -eigenvectors in the corner-pair 2-blocks for the pair X .
4. E_1^- -preservation. For $n \geq 4$, Φ_X identifies $E_1^-|_{V_{\mu_X}}$ with $\Phi_X(V_{\mu_X}) \cap E_1^-|_{V_\lambda}$.

Lemma 3 (Phase A decomposition for λ). *For $\lambda = (4, 3, 1^q)$ at $q \geq 1$,*

$$E_{n-1}^+|_{V_\lambda} = \Phi_A(V_{\mu_A}) \oplus \Phi_B(V_{\mu_B}) \oplus \Phi_C(V_{\mu_C}) \oplus S, \quad (1)$$

where S is the span of v_T for SYTs T with $T(2, 2) = n-1$, $T(2, 3) = n$.

Proof. By Theorem 1, $E_{n-1}^+|_{V_\lambda} = R'_n V_\lambda$, and the $+q$ -eigenvectors of T_{n-1} in the seminormal basis of V_λ arise from two configurations of the letters $\{n-1, n\}$ in an SYT (see SRD-421, Lemma 3/Lemma 12 there):

- (i) $\{n-1, n\}$ at a corner pair $\{c, c'\}$ in different rows and columns, giving a non-degenerate T_{n-1} -2-block whose $+q$ -eigenvector is $\Phi_{\{c, c'\}}(v_{T'})$ for the restriction $T' = T|_{\mu_{\{c, c'\}}}$. The three corner pairs are $\{c_1, c_3\} = A$, $\{c_2, c_3\} = B$, $\{c_1, c_2\} = C$.
 - (ii) $\{n-1, n\}$ in a same-row pair, giving v_T itself as a $+q$ -eigenvector (Hoefsmit rule: $T_{n-1}v_T = qv_T$ when $n-1, n$ are in the same row). The unique same-row pair of λ is $\{(2, 2), (2, 3)\}$ at row 2.
- The four families have disjoint seminormal supports (each is characterised by where $\{n-1, n\}$ sit), so the sum is direct. The total dimension is $\sum_X f^{\mu_X} + f^{\nu_2}$, which equals the total number of $+q$ -eigenvectors of T_{n-1} in V_λ . $\square$ $\square$

Theorem 4 (μ_B : same-row dies for $(4, 2, 1^*) + \dim$ formula, SRD-421). *For $\mu_B = (4, 2, 1^{q-1})$ with $q-1 \geq 2$ (equivalently $n-2 \geq 10$, i.e. $q \geq 3$):*

1. $\dim B_{j_0(\mu_B)}^{(\mu_B)} V_{\mu_B} = 3$.
2. $B_{j_0(\mu_B)}^{(\mu_B)} V_{\mu_B} \subseteq E_1^-|_{V_{\mu_B}}$.

Theorem 5 (μ_A : same-row dies for $(3, 3, 1^*) + \dim$ formula). *For $\mu_A = (3, 3, 1^{q-1})$:*

1. (SRD-331, May-13). *The j -formula $B_{j_0(\mu_A)}^{(\mu_A)} V_{\mu_A} \subseteq E_1^-|_{V_{\mu_A}}$ holds for every $q-1 \geq 2$ (i.e. $q \geq 3$).*
2. (SRD-331, May-13.) *$\dim B_{j_0(\mu_A)}^{(\mu_A)} V_{\mu_A} = 2$ structurally for $q-1 \geq 4$ (i.e. $q \geq 5$); for $q-1 \in \{2, 3\}$ the $\dim = 2$ is established computationally (mod $P = 100\,003$, $Q = 11$).*

Theorem 6 (μ_C at sharp index: j -formula on $(3, 2, 1^*)$, May-12-today). For $\mu_C = (3, 2, 1^q)$ on $n - 2 = q + 5$ letters with $q \geq 1$ (so $n - 2 \geq 6$),

$$B_{j_0(\mu_C)}^{(\mu_C)} V_{\mu_C} \subseteq E_1^- |_{V_{\mu_C}}.$$

Theorem 7 (General same-row-dies framework, SRD-421 Theorem 14). Let $\tilde{\lambda} \vdash N$ admit a same-row pair at row i , set $\tilde{\nu}_i := \tilde{\lambda} \setminus \{(i, \tilde{\lambda}_i - 1), (i, \tilde{\lambda}_i)\}$, $\tilde{\ell} := \ell(\tilde{\lambda})$, $\tilde{j}_0 := \tilde{\ell} + 1$, and $\tilde{S}_i$ the corresponding same-row span. Assume $\tilde{\lambda}_i \geq 3$ (so $\ell(\tilde{\nu}_i) = \tilde{\ell}$) and $B_{j_0(\tilde{\nu}_i)}^{(\tilde{\nu}_i)} V_{\tilde{\nu}_i} \subseteq E_1^- |_{V_{\tilde{\nu}_i}}$ (j -formula on $\tilde{\nu}_i$). Then

$$R'_{\tilde{j}_0} R'_{\tilde{j}_0+1} \cdots R'_{N-1} \tilde{S}_i = 0.$$

Theorem 8 (Hook j -formula at all k , Shift Lemma). For every hook $\eta = (k, 1^m)$ on $k + m \geq 3$ letters with $k \geq 2$,

$$B_{j_0(\eta)}^{(\eta)} V_{\eta} \subseteq E_1^- |_{V_{\eta}}.$$

3 The four-branch reduction

We generalise the May-12-Eve3 three-branch reduction to handle four summands and one extra commutation step.

3.1 Commutation through Φ_X

Lemma 9 (Commutation lemma). For each $X \in \{A, B, C\}$ and every $1 \leq k \leq n - 1$,

$$R'_k \Phi_X(v) = (T_{k-1}+1) \Phi_X(R'^{(\mu_X)}_{k-1} v) \quad \text{for all } v \in V_{\mu_X}.$$

Here $R'^{(\mu_X)}_{k-1}$ is the R'_{k-1} operator in $H_q(S_{|\mu_X|}) = H_q(S_{n-2})$, viewed via the natural index inclusion. Iterating,

$$P_{j_0, n-1} \Phi_X(V_{\mu_X}) = (T_{j_0-1}+1)(T_{j_0}+1) \cdots (T_{n-2}+1) \Phi_X(P^{(\mu_X)}_{j_0-1, n-2} V_{\mu_X}). \quad (2)$$

Proof. The factor $(T_{k-1} + 1)$ at the leftmost end of R'_k has the largest index $k - 1 \leq n - 2$. By Theorem 2(2), all factors (T_j+1) with $j \leq n - 3$ commute through Φ_X equivariantly, so the rightmost-to-second-leftmost factors of R'_k commute through Φ_X as the operator $R'^{(\mu_X)}_{k-1} = (T_{k-2}+1)(T_{k-3}+1) \cdots (T_1+1)$ acting inside V_{μ_X} . The leftmost factor $(T_{k-1}+1)$ remains outside as $(T_{k-1}+1)$.

Iterating: for $j_0 \leq k \leq n - 1$,

$$P_{j_0, n-1} \Phi_X = R'_{j_0} R'_{j_0+1} \cdots R'_{n-1} \Phi_X = (T_{j_0-1}+1) \Phi_X R'^{(\mu_X)}_{j_0-1} R'_{j_0+1} \cdots R'_{n-1}.$$

The next R'_{j_0+1} commutes through Φ_X (peeling off $(T_{j_0}+1)$ as a left outer factor) past $R'^{(\mu_X)}_{j_0-1}$ provided $(T_{j_0}+1)$ commutes with $R'^{(\mu_X)}_{j_0-1}$ on the μ_X -side; this follows from the index gap $j_0 - (j_0 - 2) = 2$ between T_{j_0} and the largest index $j_0 - 2$ appearing inside $R'^{(\mu_X)}_{j_0-1}$. (More precisely: $R'^{(\mu_X)}_{j_0-1} = (T_{j_0-2}+1)(T_{j_0-3}+1) \cdots (T_1+1)$ has largest index $j_0 - 2$, and T_{j_0} commutes with T_j for $|j_0 - j| \geq 2$, i.e., $j \leq j_0 - 2$.) Continuing in this manner, each outer factor $(T_{k-1}+1)$ stays outside, and the inner $R'^{(\mu_X)}_{k-1}$ accumulates into $P^{(\mu_X)}_{j_0-1, n-2}$. Hence (2). $\square$

3.2 The four-branch reduction

Theorem 10 (Four-branch reduction). *For $\lambda = (4, 3, 1^q)$ at every $q \geq 1$,*

$$B_{j_0}^{(\lambda)} V_\lambda = (T_{j_0-1}+1)(T_{j_0}+1) \cdots (T_{n-2}+1) \left[\sum_{X \in \{A, B, C\}} \Phi_X(P_{j_0-1, n-2}^{(\mu_X)} V_{\mu_X}) \right] + P_{j_0, n-1} S. \quad (3)$$

Proof. By definition, $B_{j_0}^{(\lambda)} V_\lambda = P_{j_0, n} V_\lambda = P_{j_0, n-1} R'_n V_\lambda$, and by Theorem 1 together with the Phase A decomposition (Lemma 3),

$$R'_n V_\lambda = E_{n-1}^+|_{V_\lambda} = \Phi_A(V_{\mu_A}) \oplus \Phi_B(V_{\mu_B}) \oplus \Phi_C(V_{\mu_C}) \oplus S.$$

Applying $P_{j_0, n-1}$ and using linearity together with Lemma 9 on each Φ_X -summand gives (3). $\square$

4 Each piece evaluated

4.1 μ_A piece: $\dim B^{(\mu_A)} = 2$ via SRD-331

By Theorem 5,

$$P_{j_0-1, n-2}^{(\mu_A)} V_{\mu_A} = B_{j_0(\mu_A)}^{(\mu_A)} V_{\mu_A} \subseteq E_1^-|_{V_{\mu_A}},$$

since $j_0(\mu_A) = q + 2 = j_0 - 1$ and $n - 2 = q + 5 = |\mu_A|$. The dimension is 2 (structural at $q \geq 5$, computational at $q \in \{3, 4\}$).

4.2 μ_B piece: $\dim B^{(\mu_B)} = 3$ via SRD-421

By Theorem 4,

$$P_{j_0-1, n-2}^{(\mu_B)} V_{\mu_B} = B_{j_0(\mu_B)}^{(\mu_B)} V_{\mu_B} \subseteq E_1^-|_{V_{\mu_B}},$$

since $j_0(\mu_B) = q + 2 = j_0 - 1$ and $n - 2 = q + 5 = |\mu_B|$. The dimension is 3 (structural for all $q \geq 3$).

4.3 μ_C piece is zero by j -formula leakage

Lemma 11 (μ_C piece vanishes). *For $q \geq 1$, $P_{j_0-1, n-2}^{(\mu_C)} V_{\mu_C} = 0$.*

Proof. Since $\ell(\mu_C) = q + 2 = \ell(\lambda)$, the sharp index for μ_C is $j_0(\mu_C) = q + 3 = j_0(\lambda)$. Our inner iteration ranges $j_0 - 1$ to $n - 2 = q + 5$, one step beyond the sharp index of μ_C :

$$P_{j_0-1, n-2}^{(\mu_C)} V_{\mu_C} = R_{j_0-1}'^{(\mu_C)} P_{j_0, n-2}^{(\mu_C)} V_{\mu_C} = R_{q+2}'^{(\mu_C)} (B_{j_0(\mu_C)}^{(\mu_C)} V_{\mu_C}).$$

By Theorem 6, $B_{j_0(\mu_C)}^{(\mu_C)} V_{\mu_C} \subseteq E_1^-|_{V_{\mu_C}}$. The operator $R_{q+2}'^{(\mu_C)} = (T_{q+1}+1)(T_q+1) \cdots (T_2+1)(T_1+1)$ has rightmost factor (T_1+1) , which is applied first to any input vector. On E_1^- , T_1 acts as -1 , so $(T_1+1)|_{E_1^-} = 0$. Hence $R_{q+2}'^{(\mu_C)} E_1^-|_{V_{\mu_C}} = 0$, giving the claim. $\square$

4.4 S piece vanishes by the same-row-dies framework

Lemma 12 (S piece vanishes). *For $q \geq 1$, $P_{j_0, n-1} S = 0$.*

Proof. We apply Theorem 7 (general same-row-dies framework) with $\tilde{\lambda} = \lambda$ and $i = 2$:

- Same-row pair at row 2: $\lambda_2 = 3 \geq 2$ and $\lambda_3 = 1 \leq \lambda_2 - 2 = 1$. ✓
- $\lambda_2 = 3 \geq 3$, so $\ell(\nu_2) = \ell(\lambda)$ (since removing two cells $(2, 2), (2, 3)$ leaves $(2, 1)$ as the row-2 cell, preserving row 2). ✓
- $\nu_2 = (4, 1^{q+1})$ is a hook on $n - 2 = q + 5 \geq 8$ letters. By Theorem 8 (Shift Lemma all-hooks) with $k = 4$ and $m = q + 1$, $B_{j_0(\nu_2)}^{(\nu_2)} V_{\nu_2} \subseteq E_1^-|_{V_{\nu_2}}$. ✓

Hence $P_{j_0, n-1} S = R'_{j_0} R'_{j_0+1} \cdots R'_{n-1} S = 0$. □ □

5 Main theorem: j -formula and upper bound

Theorem 13 (j -formula at sharp index). *For $\lambda = (4, 3, 1^q)$ at every $q \geq 3$,*

$$B_{j_0}^{(\lambda)} V_\lambda \subseteq E_1^-|_{V_\lambda}.$$

Proof. By the four-branch reduction (Theorem 10) and Lemmas 11 and 12,

$$B_{j_0}^{(\lambda)} V_\lambda = (T_{j_0-1}+1)(T_{j_0}+1) \cdots (T_{n-2}+1) \left[\Phi_A(B_{j_0(\mu_A)}^{(\mu_A)} V_{\mu_A}) + \Phi_B(B_{j_0(\mu_B)}^{(\mu_B)} V_{\mu_B}) \right].$$

By Theorems 5(1) and 4(2) for $q \geq 3$, both inner images lie in E_1^- of the respective V_{μ_X} . By Theorem 2(4), Φ_X identifies $E_1^-|_{V_{\mu_X}}$ with $\Phi_X(V_{\mu_X}) \cap E_1^-|_{V_\lambda}$. Hence

$$\Phi_A(B_{j_0(\mu_A)}^{(\mu_A)} V_{\mu_A}) \subseteq E_1^-|_{V_\lambda}, \quad \Phi_B(B_{j_0(\mu_B)}^{(\mu_B)} V_{\mu_B}) \subseteq E_1^-|_{V_\lambda}.$$

Finally, each outer factor (T_j+1) for $j \in \{j_0-1, j_0, \dots, n-2\}$ has index $j \geq j_0-1 = q+2 \geq 5$ (since $q \geq 3$). Each such T_j commutes with T_1 (index gap $j-1 \geq 4$), so each (T_j+1) preserves $E_1^-|_{V_\lambda}$. Multiplying these together still preserves E_1^- . □ □

Corollary 14 (Rank-zero). $\Pi^{S_n}|_{V_\lambda} = B_2^{(\lambda)} V_\lambda = 0$.

Proof. From Theorem 13, $B_{j_0}^{(\lambda)} V_\lambda \subseteq E_1^-$. Iterating downward: $B_{j_0-1}^{(\lambda)} V_\lambda = R'_{j_0-1} B_{j_0}^{(\lambda)} V_\lambda \subseteq R'_{j_0-1} E_1^- = 0$, since R'_{j_0-1} ends with (T_1+1) on the right (and $j_0-1 \geq 2$). Successive iterations preserve 0, giving $B_2^{(\lambda)} V_\lambda = 0$. □ □

Theorem 15 (Upper bound). *For $\lambda = (4, 3, 1^q)$ at every $q \geq 3$,*

$$\dim B_{j_0}^{(\lambda)} V_\lambda \leq 5.$$

The upper bound is structural for $q \geq 5$ and mixed structural-computational for $q \in \{3, 4\}$.

Proof. From the proof of Theorem 13,

$$B_{j_0}^{(\lambda)} V_\lambda = (T_{j_0-1}+1) \cdots (T_{n-2}+1) \left[\Phi_A(B_{j_0(\mu_A)}^{(\mu_A)} V_{\mu_A}) + \Phi_B(B_{j_0(\mu_B)}^{(\mu_B)} V_{\mu_B}) \right].$$

Each Φ_X is injective, so $\dim \Phi_X(B_{j_0(\mu_X)}^{(\mu_X)} V_{\mu_X}) = \dim B_{j_0(\mu_X)}^{(\mu_X)} V_{\mu_X}$. The sum of dimensions in the bracket is therefore at most

$$\dim B_{j_0(\mu_A)}^{(\mu_A)} V_{\mu_A} + \dim B_{j_0(\mu_B)}^{(\mu_B)} V_{\mu_B} = 2 + 3 = 5,$$

using Theorems 5(2) and 4(1). Multiplying by the outer factors (T_j+1) cannot increase the dimension. Hence $\dim B_{j_0}^{(\lambda)} V_\lambda \leq 5$.

For $q \geq 5$, both summand dimensions are structural; for $q \in \{3, 4\}$, $\dim B^{(\mu_A)} = 2$ at $\mu_A = (3, 3, 1^{q-1})$ with $q-1 \in \{2, 3\}$ is computational (SRD-331, Remark 5). $\square$ $\square$

6 Computational verification

Direct dim test for $\lambda = (4, 3, 1^q)$. Using `compute_dim.B.py` (May-13 multi-corner paper's script), mod $P = 100\,003$ at $q_{\text{spec}} = 11$:

λ	$\dim V_\lambda$	$\dim B_{j_0}^{(\lambda)} V_\lambda$
$(4, 3, 1, 1, 1) \quad (q = 3, n = 10)$	525	5
$(4, 3, 1, 1, 1, 1) \quad (q = 4, n = 11)$	1100	5

Both match the conjectural $f^{(3,2)} = 5$, agreeing with the structural upper bound and the May-13 multi-corner closed form.

Same-row part dimension. For the S -piece: $\dim S = f^{\nu_2} = f^{(4, 1^{q+1})} = \binom{q+4}{3}$. At $q = 3$: $\dim S = 35$. At $q = 4$: $\dim S = 56$. Despite this large dimension, Lemma 12 forces $P_{j_0, n-1} S = 0$.

7 Open problems and follow-up

Structural lower bound: $\dim B \geq 5$. The upper bound $\dim B \leq 5$ is matched by the computational lower bound $\dim B \geq 5$ at $q \in \{3, 4\}$. A structural lower bound would construct 5 explicit vectors $F_{X,Y}$ in $B_{j_0}^{(\lambda)} V_\lambda$ (indexed by an SYT of μ_A or μ_B combined with a corner-pair label) and separate them via position-of- n / position-of- $(n-1)$ projections, analogous to the May-12-Eve3 lower-bound argument for $(4, 2, 1^*)$. We defer this to a follow-up paper.

Dim formula at $q \in \{3, 4\}$, fully structural. Theorem 15's upper bound at $q \in \{3, 4\}$ uses the computational $\dim = 2$ for $\mu_A = (3, 3, 1^{q-1})$ at $q-1 \in \{2, 3\}$. A structural lower bound on μ_A (currently open below $q-1 = 4$) would propagate the structural upper bound on λ to $q = 3, 4$ as well.

Beyond $(4, 3, 1^*)$: the $(k, 3, 1^*)$ family. The proof of Lemma 12 (same-row-dies for λ) applies to any $(k, 3, 1^q)$ at $k \geq 4$ with $\lambda_2 = 3$ and $\lambda_3 = 1$: the sub-shape $\nu_2 = (k, 1^{q+1})$ is a hook and Theorem 8 applies. The four-branch reduction (Theorem 10) and the μ_X identifications analogously give the upper bound $\dim B_{j_0}^{(\lambda)} V_\lambda \leq f^{(k-1, 2)}$ *provided* the recursive sub-shape inputs ($\mu_A = (k-1, 3, 1^{q-1})$, $\mu_B = (k, 2, 1^{q-1})$, $\mu_C = (k-1, 2, 1^q)$) themselves have known dim/ j -formulas. For $k = 5$: $\mu_A = (4, 3, 1^{q-1})$ is the family of this paper (at one-smaller q), giving a recursive bootstrap upward.

The general program. Combining the same-row-dies framework with the multi-corner recursion gives a roadmap for proving the closed form $\dim B_{j_0}^{(\lambda)} V_\lambda = f^{\lambda(\geq 2)}$ structurally across the stable regime: each step adds a new shape class to the proven j -formulas, which in turn unlocks new sub-shape inputs via the same-row-dies framework. This paper closes $(4, 3, 1^*)$; the next natural targets are $(5, 3, 1^*)$ and $(5, 4, 1^*)$, which appear within reach using the same toolkit.

8 Honest status

What we proved. For $\lambda = (4, 3, 1^q)$ at every $q \geq 3$:

1. The j -formula at sharp index: $B_{j_0}^{(\lambda)} V_\lambda \subseteq E_1^-|_{V_\lambda}$, hence rank-zero $\Pi^{S_n}|_{V_\lambda} = 0$ (Theorem 13, Corollary 14).
2. The upper bound $\dim B_{j_0}^{(\lambda)} V_\lambda \leq 5$ (Theorem 15), structurally for $q \geq 5$ and mixed structural-computational for $q \in \{3, 4\}$.

What we did not prove. The matching structural lower bound $\dim \geq 5$. Computationally, $\dim = 5$ at $q \in \{3, 4\}$ from `compute_dim.B.py`.

What this closes. This is the first $r = 2$ same-row family beyond $(4, 2, 1^*)$ where the structural j -formula at sharp index is established. It demonstrates that the May-13 same-row-dies framework, combined with the prior j -formulas on hooks and small non-hook shapes, propagates across an infinite- q family. The proof method is robust and points toward $(5, 3, 1^*)$, $(k, 3, 1^*)$, and recursive bootstrapping up the multi-corner ladder.